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Experimental report

Experiment: Modularity of TOUCANs

Date: 08.12.2021

By: Baris Ata Borsa

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Experimental Conditions

Recording Report (96-Well Plate Reader)

Experiment Overview:

- **Study Title:** Modularity of TOUCANs
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- **Experiment Date and time:** 08.12.2021
- **Laboratory:** Microbiology Lab at 2E:686
- **Type of Lab:** BSL-2
- **Lab Technician:** Baris Ata Borsa
- **Methodology:** Broth microdilution (BMD)
- **Measurement Type:** Naked eye and OD600 measurements
- **Protocol:** SOP-TL-AST

Other information: The aim of the experiment is to show the importance of the localization of threee moieties (resistant moiety, specific moiety, drug moiety) for TOUCAN approach.

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Sample Type:

- **Sample 1:** *Staphylococcus aureus* ATCC29213

Drug and concentrations used:

- RSD (TOUCAN-MRSA): mU-mU-mU-mU-T-T-Fx
- SRD: T-T-mU-mU-mU-mU-Fx
- RDS: mU-mU-mU-mU-Fx-T-T
- DRS: Fx-mU-mU-mU-mU-T-T
- SDR: T-T-Fx-mU-mU-mU-mU
- DSR: Fx-T-T-mU-mU-mU-mU
- RRD: mU-mU-mU-mU-Fx

➢ Final concentration was 40 µg/ml for all tested drugs.

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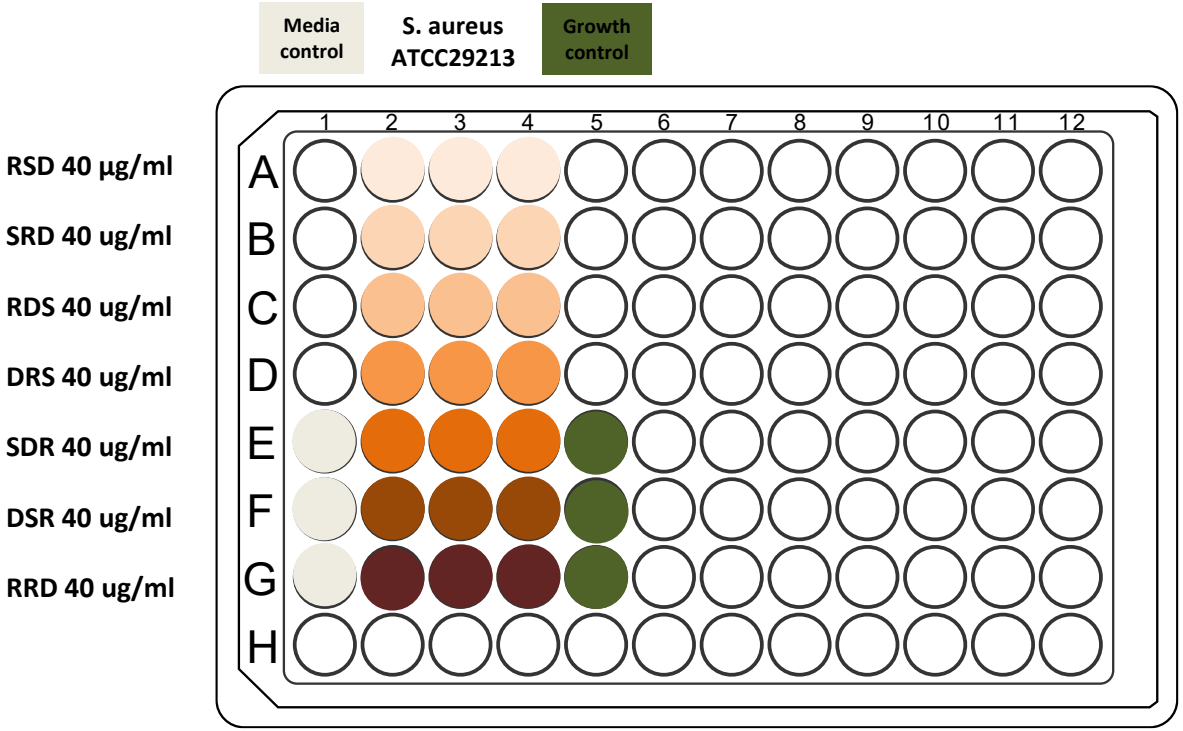
Experiment design

Plate Layout (96-Well Format):



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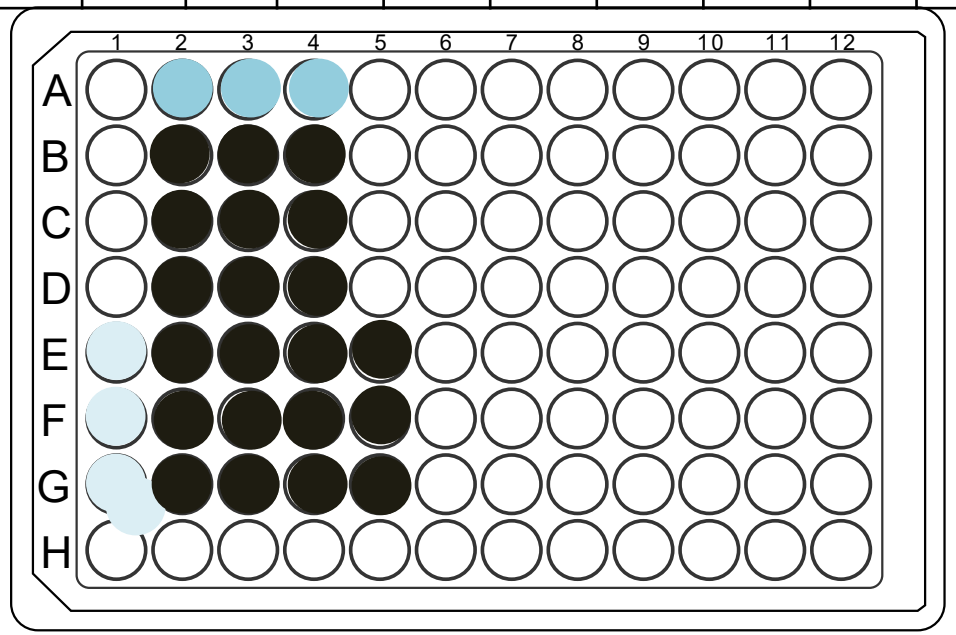
Data acquisition

Raw Data: OD600 at 18h

	Media control	S. aureus ATCC29213			Growth control											
RSD 40 µg/ml		0,207*	0,211*	0,236*												
SRD 40 ug/ml		1,429	1,477	1,498												
RDS 40 ug/ml		1,423	1,464	1,462												
DRS 40 ug/ml		1,469	1,516	1,503												
SDR 40 ug/ml	0,086	1,395	1,417	1,455	1,533											
DSR 40 ug/ml	0,086	1,509	1,522	1,573	1,572											
RRD 40 ug/ml	0,085	1,587	1,584	1,595	1,595											

**See the notes and color codes below..*

Raw Data: Naked eye inspection



Growth

No Growth

Notes:

Limited growth was observed (85,3% inhibition by OD600).
This type of growth ("tiny buttons") is disregarded for Gram-positive bacteria with bacteriostatic antibiotics based on EUCAST recommendations (see below).

https://www.eucast.org/fileadmin/src/media/PDFs/EUCAST_files/MIC_testing/Reading_guide_BMD_v_5.0_2024.pdf

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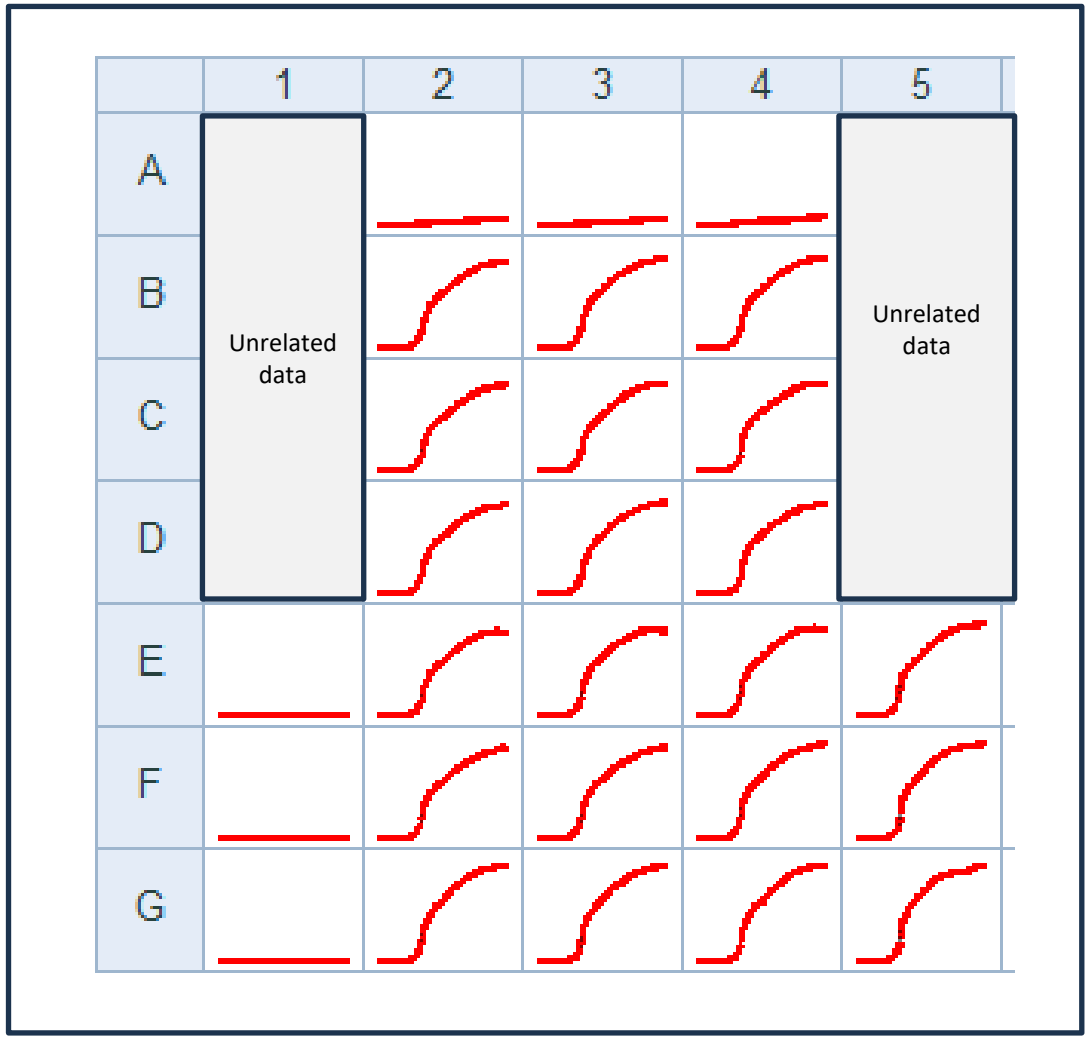
Data acquisition

Raw Data:
Growth curves from the device.



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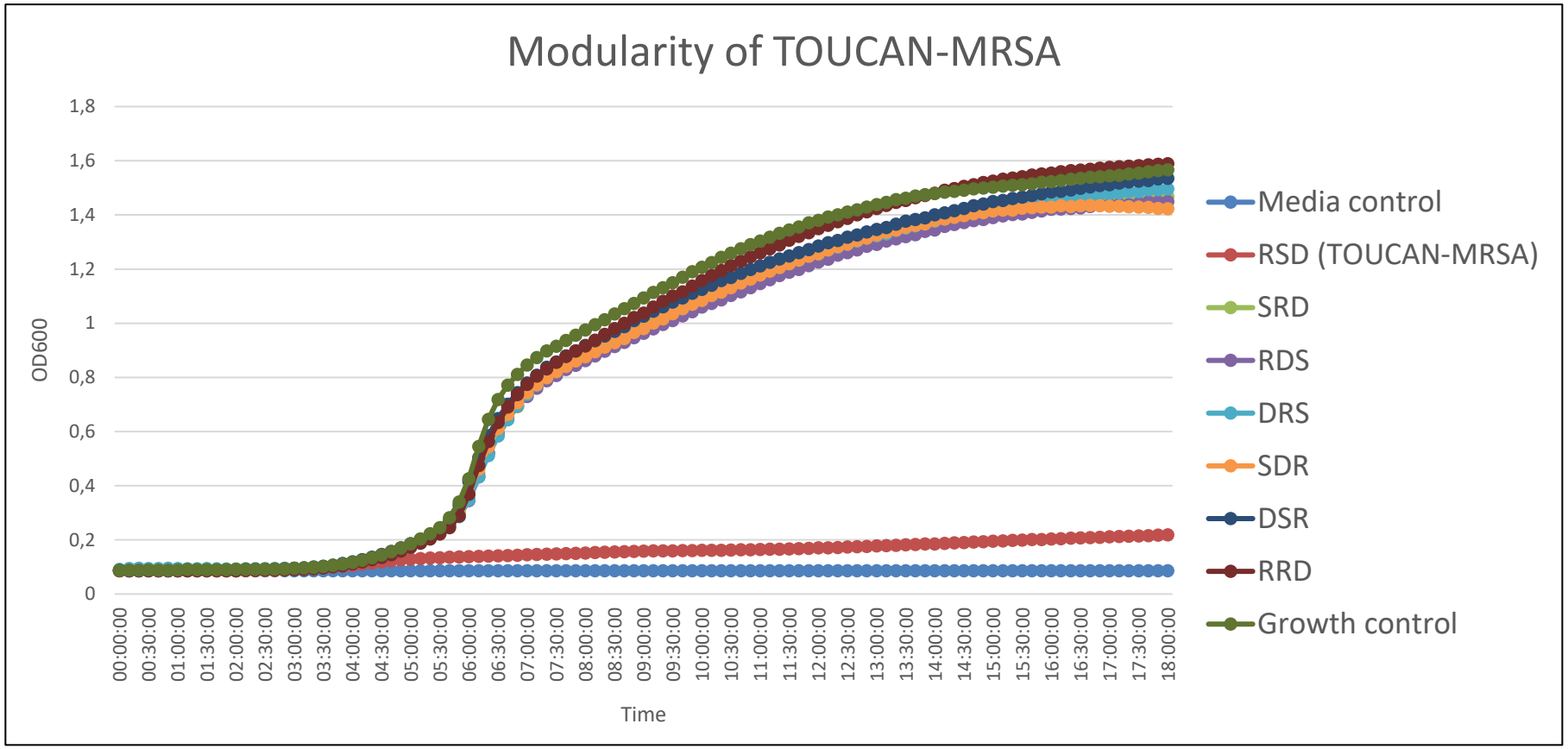
Data visualization

- Graphic for **average OD600 values over 18h** are presented below.



Date: 09.12.2021
Signature

Researcher:
Baris Ata Borsa

Date:
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Date:
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